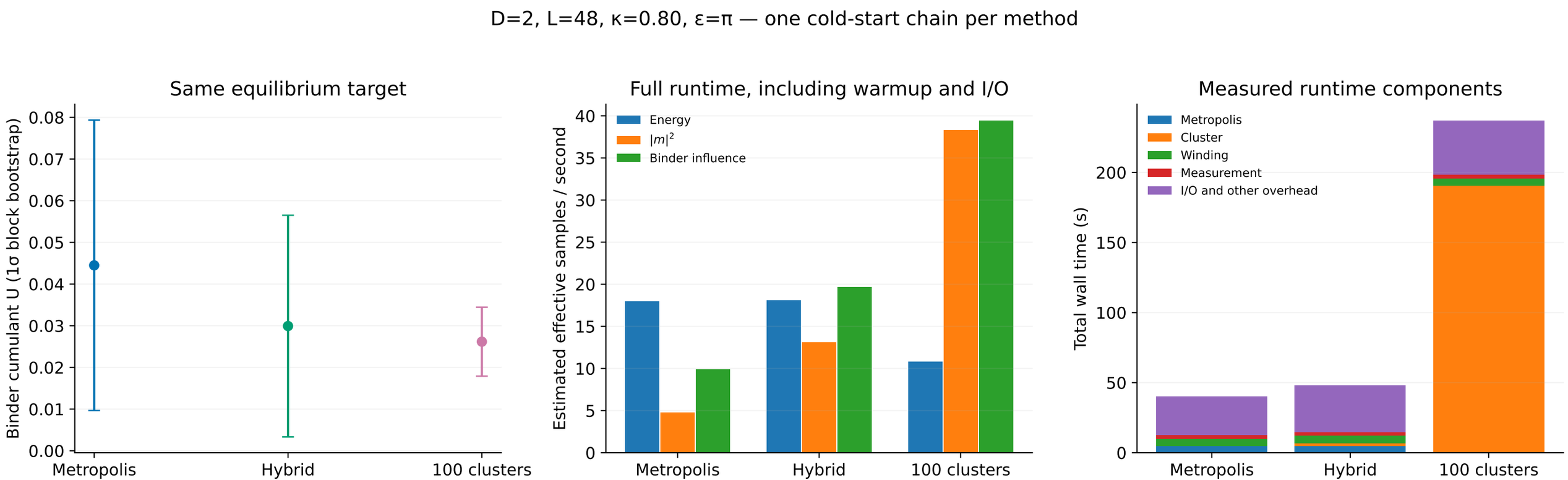
D=2, L=48, κ=0.80, ε=π — one cold-start chain per method
Same equilibrium target
Binder cumulant U (1σ block bootstrap)
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
Metropolis
Hybrid
100 clusters
Full runtime, including warmup and I/O
Estimated effective samples / second
0
5
10
15
20
25
30
35
40
Energy
$|m|^2$
Binder influence
Metropolis
Hybrid
100 clusters
Measured runtime components
Total wall time (s)
0
50
100
150
200
Metropolis
Cluster
Winding
Measurement
I/O and other overhead
Metropolis
Hybrid
100 clusters
10,000 fixed cycles per method; discard first 500; all methods add one winding proposal per cycle